Quantum optimisation for heterogeneous mesh partitioning: an open, exactly graded benchmark of QAOA and varQITE

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Abstract. Parallel PDE solvers split a mesh between compute devices; on heterogeneous clusters each device should receive work in proportion to its speed while the cut between parts stays short. We cast this two-device split with a target work share α as an Ising problem and benchmark quantum samplers against classical heuristics on nine reproducible finite-element meshes (12–24 coarse regions, one qubit each), grading every method against the exact optimum. In noiseless simulation, QAOA beats random sampling (3.9× per shot at α=0.5; 1.9× the fair target-share sampler at α=0.8) and matches local search at 12 qubits, but falls far behind classical methods at 16–20 qubits. IonQ's varQITE reaches a near-optimal split on 78% of shots at α=0.5, yet fails at α=0.8 because of a structural limit of its ansatz that we identify. We claim no quantum advantage.

1 Introduction

Graph partitioning of finite-element meshes decides load balance and communication in parallel solvers. IonQ with Ansys/Synopsys recently inserted quantum graph partitioning into LS-DYNA to reduce fill-in of sparse direct solvers, using varQITE [1] and Iterative-QAOA [2], always with *balanced* halves and closed code; quantum annealing has been tried for HPC load balancing [3]. We study the *heterogeneous* case (unequal device speeds, hence unequal target shares), release all code, and grade every quantum and classical method against exact enumeration on the same objective.

2 Problem, benchmark and model

Benchmark. Nine plates meshed with Delaunay triangulation (1 670–2 166 triangles): a rectangle with a known answer, an L-plate with a heat source, two sources, a plate with a hole (stress-style mesh), and a five-step complexity ladder on one L-plate (uniform → three sources → heavy-material zone with 3× work → strong-coupling zone with 3× border cost → priority zone). Each mesh is grouped into n connected regions by k-means with a connectivity repair; region work and border cost are summed exactly.

Ising model. With z_i=+1 (device A) or −1 (device B), normalised work $\tilde{a}_i$ and border costs $\tilde{c}_{ij}$,

$$E(z) = \sum_{\langle ij \rangle} \tilde{c}_{ij} \tfrac{1-z_i z_j}{2} + \lambda \Big(\sum_i \tilde{a}_i z_i - (1-2\alpha) \Big)^2 \; [+\mu P(z)],$$

following the Ising form of graph partitioning [4]. The balance term couples *every* pair of regions ($n(n-1)/2$ two-qubit terms), which suits IonQ's all-to-all connectivity. Scenarios: two equal devices (α=0.5, λ=10) and CPU+GPU with an assumed 4× faster GPU (α=0.8, λ=3.98); λ is shared per scenario and selected by exact enumeration. A synthetic 8-device pool (CPUs/GPUs with speed, memory and power) shows the device-choice layer: fastest plan 9.4 h/27.3 kWh, overnight 11.8 h/23.7 kWh, lowest energy 13.4 h/22.0 kWh, versus 37.5 h/56.1 kWh for equal shares (modelled, not measured).

3 Methods

Quantum (noiseless statevector, cross-checked with Qiskit to 10^{-15}). QAOA with cost layer $e^{-i\gamma E}$ and mixer $e^{+i\beta \sum X}$: linear-ramp LR-QAOA [5]; optimised angles with COBYLA, Nelder–Mead, Powell, L-BFGS-B (budget 200 evaluations) and shot-based SPSA [6]; CVaR-QAOA [7]; warm starts [8]; a symmetry-reduced variant. varQITE [9]: each step solves $G\dot{\theta} = d$ with $d_k = -(\langle P_k H\rangle - \langle P_k\rangle\langle H\rangle)$ and $G_{kj}{=}\partial\langle P_k\rangle/\partial\theta_j$ by parameter shift, for the paper's $e^{i\theta Z_r Y_q}$ ansatz on all pairs (ZY), a R_Y+CX ansatz on all pairs or graph edges (A), our extension with one R_Y per qubit (A$^+$) and an IonQ-native RZZ+RX ansatz (B), all trained on exact expectations. **Classical.** Uniform and target-share random sampling, simulated annealing, local search with restarts, spectral bisection; same objective, same budget of 1 000 candidates as QAOA's 1 000 shots. **Honesty measures.** All schedules, step sizes and annealing temperatures are tuned on separate calibration meshes, never on scored instances; the p=0 control and the target-share sampler are always reported; the main metric is the probability of a split with cost ≤ 1.05× optimum; 30 injected bugs (signs, bit order, schedules, metric formulas) are all caught by the cell's checks; a crude depolarising estimate $F{=}(1{-}0.005)^{N_{2q}}$ indicates noise sensitivity.

4 Results

12 qubits (Table 1). All QAOA variants beat random sampling on every case: LR-QAOA raises the per-shot success 3.9× at α=0.5; at α=0.8 it is 22× uniform guessing but only 1.9× the classical target-share sampler (QAOA+COBYLA 2.6×). An ablation explains the gap: the single-region terms alone give 0.30% per shot, the pair terms alone 0.01%, both 0.78%. With 1 000 shots QAOA matches the best classical heuristic (local search 86%, QAOA+COBYLA 92% at α=0.5), but at 4 096 states exhaustive search is exact and cheap. Warm starts inherit most of their success from the spectral split; the convex-relaxation warm start is degenerate here ($c_i{=}\alpha$ for every region).

Scaling (Fig. 1a,b; 216 instances, 2 928 QAOA runs on a Hetzner node). At fixed depth p=2 QAOA loses about an order of magnitude per shot for every four extra qubits. At 20 qubits QAOA+COBYLA finds a split within 5% in 2% / 21% of runs (α=0.5 / 0.8), versus 50% / 100% for spectral bisection and 18% / 77% for local search.

varQITE (Fig. 1c,d). At α=0.5 the paper's ZY ansatz converges to near-optimal splits (78% per shot, 30–100% on every case) with 66 two-qubit gates, but only after ≈ 6 473 exact-expectation training circuits. At α=0.8 every varQITE variant fails: in ansatz A and ZY a qubit that is only ever a control keeps a 50/50 marginal for all angles, which caps a unique optimum at probability 1/2; A$^+$ removes the cap and reaches $r{\approx}0.98$ but still misses the 5% window. Graph-edge versions (22 gates) are too weak.

Noise and outcome. The depolarising estimate lowers LR-QAOA (p=2, α=0.8) from 0.78% to 0.42% per

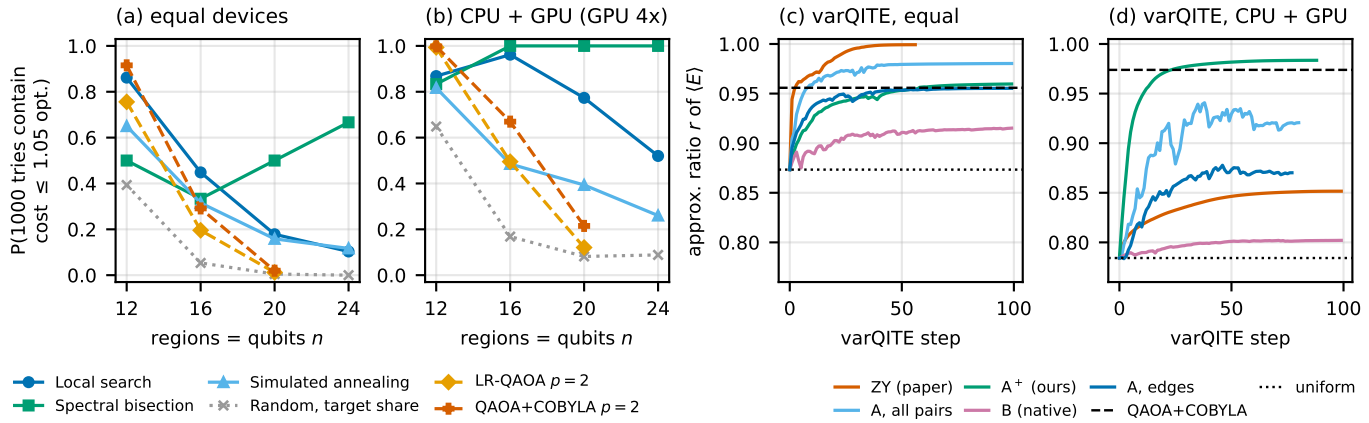


Figure 1: (a,b) Probability that 1 000 candidates (classical) or shots (QAOA, noiseless) contain a split within 5% of the optimum, versus problem size (cases 2 and 9, seeds 7–9; the 24-qubit reference is the best classical result; QAOA is not simulated at 24 qubits). (c,d) varQITE training curves at n=12 (mean over nine cases), with QAOA+COBYLA and uniform sampling for reference.

Table 1: Results at n=12 qubits (nine cases, noiseless): two-qubit gates, training circuits, per-shot probability of cost $\leq 1.05\times$ optimum (%), and probability that 1 000 shots/candidates contain such a split (%). †Starts from the classical spectral split.

			equal devices (α=0.5)		CPU+GPU (α=0.8)	
Method	2q gates	training circuits	per shot	hit@1000	per shot	hit@1000
Uniform random sampling	0	0	0.07	48	0.04	29
Random with target share	0	0	0.07	48	0.41	74
Simulated annealing	0	0	–	64	–	87
Local search (restarts)	0	0	–	84	–	100
Spectral bisection	0	0	–	67	–	100
LR-QAOA, $p = 2$	132	0	0.27	85	0.78	99
QAOA + COBYLA, $p = 2$	132	200	0.36	94	1.1	100
CVaR-QAOA + COBYLA, $p = 2$	132	31	0.31	90	0.87	99
Warm start (spectral)†, $p = 2$	132	200	7.8	94	21	100
varQITE ZY (paper, all pairs)	66	8 675	78	100	<0.01	<0.01
varQITE A (all pairs)	66	7 566	24	33	<0.01	0.01
varQITE A$^+$ (edges, our ext.)	22	5 003	<0.01	<0.01	<0.01	<0.01
varQITE B (IonQ-native, edges)	22	4 686	0.01	13	0.01	7.8

shot and makes deeper circuits worse. The modelled runtime of every method's best split lies within 2.9% of the runtime-optimal split.

5 Discussion and limitations

The pipeline works end to end and quantifies where today's quantum samplers stand on a real engineering objective: competitive at 12 qubits, outpaced by classical heuristics beyond, and—for varQITE—strongly dependent on ansatz structure under unequal shares. All results are noiseless simulations or models; varQITE training used exact expectations; runtimes, energies and device speeds are modelled assumptions. At 16 qubits the graph-edge varQITE variants still never reach the 5% window. Four of the nine meshes (cases 1, 2, 4, 5) differ at bit level between the macOS/arm64 and Linux/x86-64 builds, so the server's large-mode case-2 instances are equivalent but not identical to the notebook's; shipping meshes as data files will fix this. Next: the single batched IonQ Forte 1 noise-model run (p=0 control, QAOA and the final trained varQITE circuits), and measured CPU/GPU runs of refined meshes on HPC.

This effort is supported by Qollab & IonQ. Code and all result files: `https://github.com/Oussema-t/zollerqu-qollab` (branch `mesh`). AI assistants (Claude Code, ChatGPT) were used for code and text; every number is produced by the published code.

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